SIOP 2026 · ML COMPETITION

ONE HOT KEY

ONE PERSON + AGENTS vs HUMAN TEAMS

Bivariate Pearson r extraction · IO Psychology meta-analysis

Team: One Hot Key · Task: Hidden-Set MSE · Tool: Walt (RTX PRO 3000 Blackwell)

PROJECT OVERVIEW

The Challenge

Build an automated pipeline that reads academic PDFs and extracts **zero-order bivariate Pearson r correlations** for meta-analytic synthesis in industrial-organizational psychology.

Dev set

127 studies on trust × subjective well-being

Test set

23 distinct construct pairs across 66 papers — out-of-distribution

0.013641

BEST DEV SET MSE

127 + 66

STUDIES PROCESSED

23

CONSTRUCT PAIRS (TEST)

SCORED ON **Mean Squared Error vs. hidden ground-truth r values**

SIX GATES, CASCADING FALLBACK

Pipeline Architecture

Each PDF passes through a cascade — earlier gates are cheap and high-precision, later gates are expensive and recover what slips through.

01 **PDF Acquisition** →
02 **Layout Extraction** →
03 **Regex + phi4 Classifier** →
04 **Vision Fallback** →
05 **Structured LLM Extraction** →
06 **Imputation**

EARLY EXIT: if a high-confidence value is found at any gate, downstream gates are skipped for that pair.

GATE 01 · PIPELINE STAGE

PDF Acquisition

Robust DOI-to-PDF resolution with a four-stage fallback chain

batch_pdf_agent.py walks each study's DOI through progressively more aggressive resolvers, saving as study1.pdf … study127.pdf with a download_log.json audit trail.

- CrossRef → publisher metadata and direct PDF links
- Unpaywall → open-access copies of paywalled articles
- Semantic Scholar → corpus-level fallback when the above miss
- Playwright → headless-browser scrape for HTML-gated PDFs

Playwright

PYTHON PACKAGE

ACQUISITION RATE **193/193 PDFs successfully retrieved across dev + test sets**

Layout-Aware Extraction

Two-tier table extraction — geometric first, ML for the hard cases

pdfplumber runs first (Tier 0): geometric table detection with cell-bbox metadata, fast and exact on regular layouts. Docling (Tier 1) takes the residual — multi-column papers, rotated pages, combined headers — using ML TableFormer to recover structure where geometry alone fails. PyMuPDF / fitz is the underlying PDF reader for both.

- pdfplumber: geometric tables with proximity search across rows × columns
- Docling: ML TableFormer for complex / combined-cell layouts, optionally cross-validated by qwen2.5vl on the table crop
- Cell adjacency preserved end-to-end so downstream variable lookup works
- Falls through to vision (Gate 4) only when both layout parsers come up empty

pdfplumber + docling

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PRIMARY USE Correlation matrices, descriptive statistics tables, MBI subscale tables

GATE 03 · PIPELINE STAGE

Regex + phi4 Classification

Pattern-matching candidate r values, then LLM-classifying which one matches the construct pair

Two-stage filter. Construct-aware regex walks PyMuPDF-extracted text and table cells to gather every plausible r candidate with its surrounding context. phi4 (via Ollama) then reads each candidate against construct definitions and decides which row × column pair the value belongs to — the role no rule-based pattern can do reliably.

- Regex pass: distinguishes Pearson r from Spearman ρ, betas, and probit coefficients
- Rejects ANOVA, imputation, and cross-wave correlation tables explicitly
- phi4: classifies candidates against construct definitions, with same-wave longitudinal rule and inverse-label sign-flip logic in-prompt
- Handles Unicode significance markers (∗ U+2217, ⁎ U+204E) vs. ASCII *

Python re + phi4

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PRECISION FOCUS False positives cost as much as misses — phi4's job is rejecting wrong-table matches the regex couldn't filter

GATE 04 · PIPELINE STAGE

Vision Fallback

Visual table parsing for cases where layout extraction fails

When pdfplumber and Docling both come up empty — typically scanned PDFs, image-embedded tables, or unusual layouts — the pipeline falls back to a local vision-language model. PyMuPDF rasterizes the relevant page; qwen2.5-VL reads the image directly and returns structured cells.

- qwen2.5-VL:7B reads the rendered page image and emits structured JSON cells
- Page rasterization handled by PyMuPDF (fitz) at the resolution the model needs
- Runs entirely local — no API cost, no rate limits, full offline reproducibility
- Construct classification and sign-flip logic still applied post-hoc by Gate 3 logic

Qwen2.5-VL

via Ollama

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WHEN IT FIRES **Scanned PDFs, image-only tables, papers Docling parses but where text-layer extraction yields no candidates**

Structured LLM Extraction

How the hardest cases get extracted — process, not vendor

A separate module — pipeline_v9_opus_sweep.py — runs a custom extraction loop on a curated short-list of studies the open-source pipeline missed or low-confidenced. The model behind the call is interchangeable; the methodology is what makes the values trustworthy.

- PROMPT: construct definitions, sign-flip rules, same-wave longitudinal rule, and explicit rejection criteria for ANOVA / regression beta / cross-wave traps
- INPUT: full PDF + a list of which (X, Y) construct pair to extract for that paper, plus measure aliases pulled from each paper's methods section
- OUTPUT: JSON-schema-constrained — { r, n, page, table_cell, confidence, supporting_quote } — parsed and type-validated before acceptance
- AUDIT: every accepted value carries its supporting quote, so the manifest can be spot-checked by re-reading the cited sentence rather than re-running the model

Schema-constrained sweep

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WHY THIS GATE EXISTS **To recover the ~10% of correlations that survive bad layout, sign-flips, or table-cell ambiguity that defeats every upstream gate**

GATE 06 · PIPELINE STAGE

Imputation

Last-resort fill for studies where every gate above missed

When all five extraction gates return nothing, the pipeline imputes a single value chosen to minimize MSE under uncertainty. The optimal value is back-calculated from paired dev-set submissions.

- Optimal imputation value ≈ 0.152 (data-driven, not a heuristic)
- Back-calculated by submitting paired dev runs and solving for the minimum
- U-curve flat at optimum — further imputation tuning yields no gain
- Coverage at upstream gates is the only remaining lever for improvement

pandas · numpy

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KEY INSIGHT **Imputation is not a differentiator. Pipeline coverage is.**

Division of Labor

Matt

Strategic lead

Pipeline strategy, GT coding, PDF reading, batch-log diagnosis, prompt-writing for Cursor.

Claude

Diagnostician

PDF inspection, batch failure analysis, precise Cursor-prompt drafts, file-validation scripts.

Cursor

Implementer

All code reading, concrete bug-finding, and pipeline implementation. No architectural decisions made unilaterally.

PowerShell

Runtime

Batch runs and spot checks only — never used as an iterative debugging surface.

GOLDEN RULE Diagnose once, write one comprehensive prompt, let Cursor implement. Two cycles maximum.

DEV → TEST OUT-OF-DISTRIBUTION

Validation Strategy

No traditional k-fold CV — the competition provides a fixed dev/test split with deliberate distribution shift in construct space.

DEV SET

127

studies on trust × subjective well-being

- Held-out ground-truth manifest manually coded
- 57 numeric, 8 confirmed BLANK, 1 PENDING
- MSE used as fast iteration signal

TEST SET

66

papers across 23 distinct construct pairs

- Generalizes well beyond trust × wellbeing
- Forces removal of all study-specific hardcodes
- Hidden-set MSE — only visible after submission

PAIRED-SUBMISSION TRICK

Optimal imputation value back-calculated by submitting matched dev runs and solving the resulting two-equation system. The U-curve was confirmed flat at the optimum, ruling out further imputation tuning as a path forward.

DEV vs. TEST

Results

DEV SET

0.0136

Best MSE · submission_v11_study59

Held-out trust × wellbeing manifest
Used as fast iteration signal

TEST SET

0.0351

Best test MSE · 2026-04-11

23 construct pairs across 66 papers
Out-of-distribution generalization

WHAT CLOSED THE GAP

Removing study-specific hardcodes from v11, generalizing construct classification to 23 pairs, and recovering false-negative regex matches on high-r studies (study41 = 0.73, study17 = 0.65, study56 = −0.57).

ONE PERSON + AGENTS vs HUMAN TEAMS

Dev Set Leaderboard

How a one-person-plus-agents experiment landed against teams of researchers and analysts on the dev set.

RANK	TEAM	ERROR ↓	SUBMITTED
1	Hungry Llama	0.006336	2026-04-06
2	MetaML	0.009370	2026-03-25
3	Putka's Army	0.011521	2026-03-29
4	ILMNinsights	0.013198	2026-03-26
5	STATS Lab at CMC	0.013510	2026-04-01
6	***One Hot Key***	**0.013641**	2026-04-05
7	DSTST	0.015793	2026-04-08
8	Cam the Ram	0.026518	2026-03-23
9	Chen-Chillas	0.029143	2026-04-10
10	Neural Nomads	0.054047	2026-04-08

6 / 10

Dev set rank

THE ASYMMETRY

1 developer + Claude + Cursor.

Within 0.0005 MSE of the lab and the squad ranked above. Test set ranking pending close of competition.

Limitations

WORKSTATION · Walt

Core Ultra 9 285HX · 24 cores (8P+16E) / 24 threads · **RTX PRO 3000 Blackwell** · 12 GB GDDR7 · **64 GB DDR5**

12 GB VRAM forces phi4 over the bigger model

qwen2.5:72b — Ollama's strongest open-weights extraction tier — needs ~40 GB VRAM at Q4, over 3× what the RTX PRO 3000 Blackwell carries. Phi4 (~14B, ~9 GB Q4) is the largest model that stays fully GPU-resident; the 72B class would spill to CPU and lose roughly an order of magnitude in throughput.

Single-machine sequential

All 193 PDFs run through six gates on one workstation. Cluster-scale competitors run dozens of pipeline variants in parallel; we run one full sweep at a time, which gates how many ideas we can test per day.

Ollama serializes inference

24 cores and 992 AI TOPS, but Ollama runs one vision query at a time per model. Effective throughput on Gate 4 is bounded by single-stream latency, not total compute — so even the model that fits is queue-bound on a 127-paper batch.

Vision gate capability ceiling

qwen2.5vl-7B fits the 12 GB budget, but the 32B-and-up vision tier that would close the scanned-table OCR gap doesn't fit without aggressive quantization that erodes the accuracy gain. Vision is the most accuracy-sensitive gate in the pipeline, and it's structurally capped one tier below state-of-the-art.

WHAT THIS PROJECT TAUGHT

Key Learnings

No study-specific hardcodes

Every fix must generalize across the 23 construct pairs. Study-specific overrides were identified and removed during the v11 → v12 migration.

Silent failures are the hardest bugs

Dynamic mode never fired because process_study skipped build_study_config; Unicode asterisks vs ASCII broke parsing in study59. Both required targeted internal-state dumps to surface. Note: IO is much more fun than CS.

False positives cost as much as misses

ANOVA tables, imputation tables, Spearman correlations, and cross-wave pairs all required explicit rejection logic. Precision is not free.

Coverage, not imputation, is the lever

Optimal imputation was back-solved analytically. The U-curve is flat. The only remaining path to a lower MSE is recovering the missed high-r studies upstream.

ML Competitions are pretty fun.